Programming and Laboratory — Mod. 2

1. Introduction

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Hello!

- I am **Giulio Ermanno** Pibiri:
  - https://jermp.github.io
  - giulioermanno.pibiri@unive.it
  - https://github.com/jermp

- Associate prof. of Computer Science at Ca' Foscari University of Venice

- Teaching activities at Ca' Foscari:
  - Introduction to Computer Programming (previous semester)
  - **Programming and Laboratory — Module 2 (this course)**
  - Algorithms for Modern Hardware (advanced algorithms course; third year)

- Research interests:
  - algorithms and data structures
  - data compression
  - recently, with applications to Bioinformatics

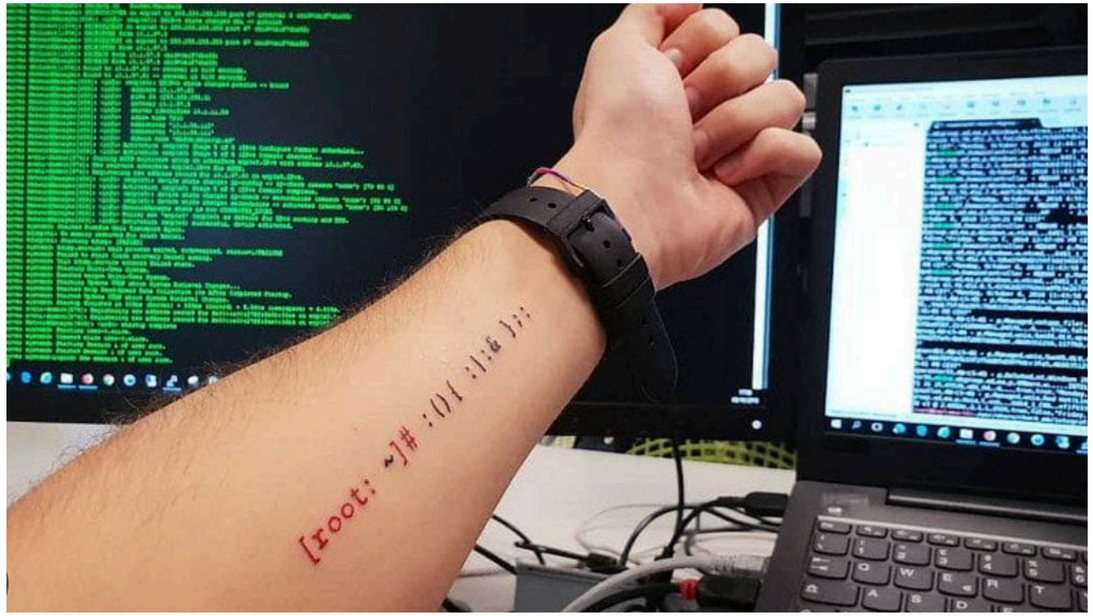
What is this course?

- Welcome to **“Programming and Laboratory — Mod. 2”**.
- This is an introductory course about C++: **fundamentals + coding** (debugging and exercises).
- C++ fundamentals:
  - generic programming
  - copy/move semantics
  - function/operator overloading
  - C++ standard template library (STL): containers, iterators, algorithms

Overview

- Course introduction and general information:
  - course structure
  - program
  - material
  - exam
- The C++ programming language
- Setting-up a development environment:
  - terminal
  - compilation

Coding



- Do the exercises!
- To be tattooed on your arm:
  practice makes you better.
- Method:

  1. Study some techniques.
  2. Read someone else's code and try to understand it.
  3. Try to write something on your own.
  4. Repeat.

- Approach: **be skeptic**. Verify everything taught in class by your own.

Program

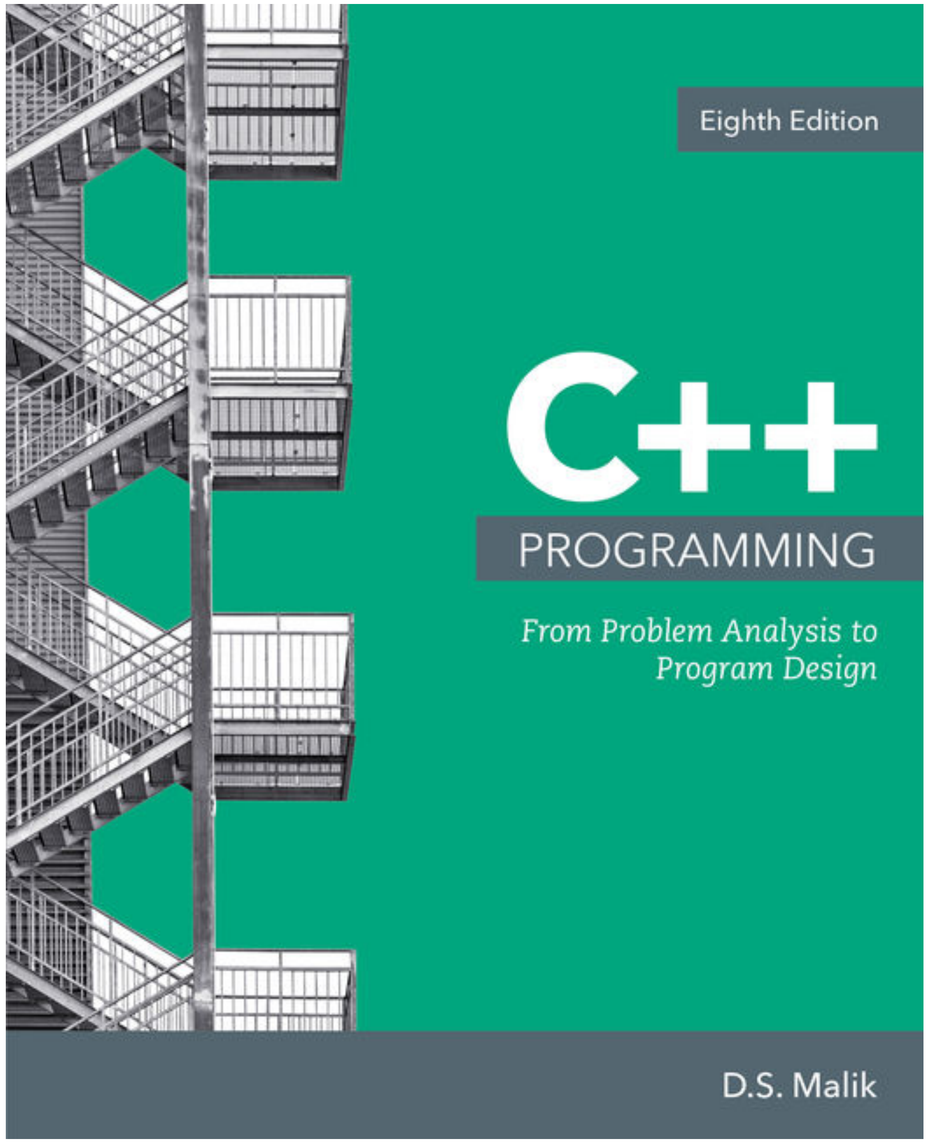
- 48h course — 6 CFU [24 lectures]
- Topics:
  1. Introduction and tools: shell, make, git, sanitizers [5 lectures]
  2. Warm-up on recursion [2 lectures]
  3. Advanced recursion: parsers and I/O streams [4 lectures]
  4. Templates [4 lectures]
  5. Containers: copy/move semantics, operator overloading, iterators [6 lectures]
  6. Standard Template Library (STL) [3 lectures]

Lectures

- Lectures are **in presence**.
- Video-recordings will be also made available for the course to better support your study. The recordings are **not** meant to replace the lecture itself.
- **IMPORTANT:** To access the recording of the lectures, you must register on Moodle using your `@stud.unive.it` account. Any other email account does not work.
- In fact, it is **very important to participate actively** during the lectures!
- All the material we will cover **is part of the exam** (written exam and project).
- We are going to meet each week:
  - on Wed, 14-15:30
  - on Thu, 12:15-13:45

Material

- Slides and code on **Moodle**; this book.



Exam workflow

Q. How to pass the exam? **Written exam** + **oral exam** + **project**.

- **Module 1:**

  1. **Written exam.**

     The first call ("appello") — **and only the first call** — for the written exam can be replaced by 4 intermediate tests, to be done in class under the supervision of **prof. Pistellato**. Only if at least 60% of the assigned exercises is sufficient the written exam can be replaced by these facultative tests.

  2. **Oral exam**. Can be done only if written exam has been passed. If you fail or do not show up, you have to **repeat** also the written exam.

- **Module 2**: **Project** (more details in next slides).

Exam rules

- Module 1 counts 70% on the final evaluation.
- Module 2 (the project) counts 30%.
- **IMPORTANT: Evaluations of Module 1 and Module 2 remain valid only for the current academic year. After the last call in January (see below) any partial grade is lost.**
- The project will be described and assigned towards the end of the course (in May).
- You will have **4** calls ("appelli") to submit the project for evaluation:
  June, July, September, and January.
- **After January, you will have to follow a new project specification (relative to the next academic year).**

Module 2 exam — individual project

- You will have to implement a **C++ container** (more details during the course) solving a non-trivial task.

- Projects of the previous years:
  - https://github.com/PEL-unive/progetto-22-23
  - https://github.com/PEL-unive/progetto-23-24
  - https://github.com/PEL-unive/project-24-25

- Remember: all implemented functions will be tested thoroughly (move/copy semantics, constructors, destructor, memory management, operator overloading, Iterators, I/O, etc).

  You will have to run thorough tests on your project!

  (Projects often fail because they have not been tested properly.)

Module 2 exam — individual project

Do not commit plagiarism!

- Every year, several students copy someone else's project (plagiarism).
- We use a **plagiarism-detector** on the projects of all sessions in the academic year.
- **We take this seriously**. What if you commit plagiarism?

Module 2 exam — individual project

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- We use a **plagiarism-detector** on the projects of all sessions in the academic year.
- **We take this seriously**. What if you commit plagiarism?

- If the plagiarism is clear:
  1. You have to **repeat the exam next year**. Module 1 is canceled, if you already passed it.
  2. Ca' Foscari's disciplinary committee is notified. You risk to be **expelled** from the university (the same happens if you copy in any other exam).
- If there is **only a suspect** of plagiarism: there will be an **oral exam on the project**.

Module 2 exam — individual project

Recommendations to avoid plagiarism:

- Discussing the high-level ideas for solving the project is ok, but **no code has to be shared** between students. If you share code, be sure that our plagiarism detector will find it out.
- **Do not post your code in a public online repository**, or in any place where other students can find it. Some other student will find it and copy it (this happened in the past).
- **Do not copy-paste code snippets** from the internet. This is also plagiarism!

Questions

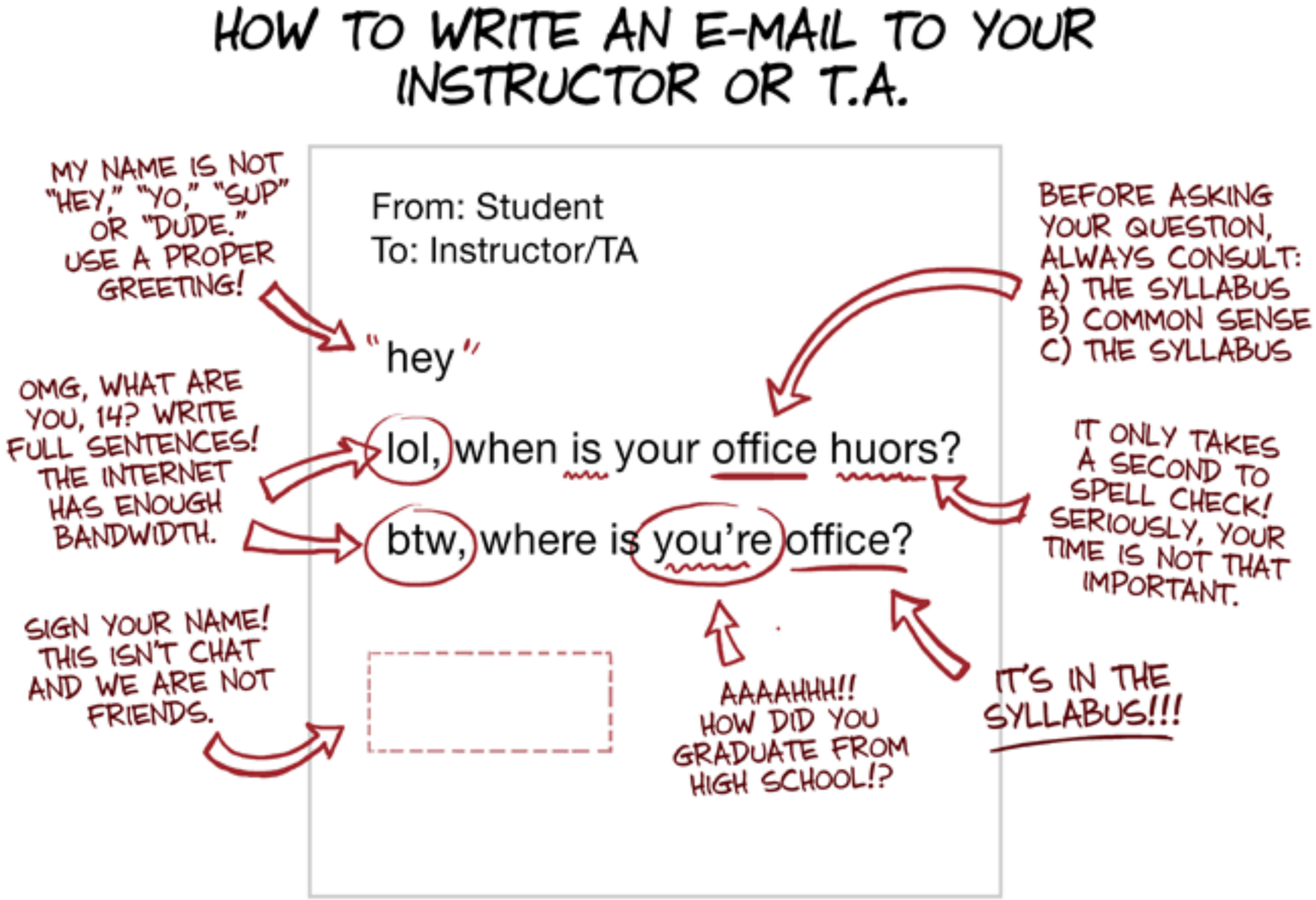
- You are kindly invited to ask questions at any time during the lectures.
- You can also ask questions via emails to our TAs:

  Davide Cologni (davide.cologni@unive.it)
  Matteo Grazioso (matteo.grazioso@unive.it)

- You can ask me for a meeting (ricevimento), either on-line or in my office.

  Send me an email to fix a date for a meeting.

E-Mails



https://marktomforde.com/academic/undergraduates/Email-Etiquette.html

E-Mails

- Please, read this Web page **carefully**:

  https://vigna.di.unimi.it/comescrivere.php

  by prof. Sebastiano Vigna (Università degli Studi di Milano).

The C++ programming language

The C++ programming language

- C++ is a **high-level**, **general-purpose**, **compiled** programming language.
- Very powerful: there is literally nothing that cannot be done in C++.
- False myth: “C++ is fast”.
  No. C++ gives you **control over performance**.

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- Originally developed by **Bjarne Stroustrup** at Bell Labs in 1983.
- At the beginning: “C with classes”.
  A lot of further development, e.g., templates, STL, etc.
- Standardized in 1998 (C++98). Following standards (major revisions):
  C++03, **C++11**, C++14, C++17, C++20.

Bjarne Stroustrup

The C++ programming language

- Since 1998, the C++ standard consists of two parts:
  - The language **core** (think of this as everything that can be written without extra libraries, with no `include` directives);
  - The standard library (stdlib), which includes most of the **Standard Template Library (STL)**.
- STL was originally a third-party library, developed by Alexander Stepanov, before it was included in the C++ standard. (More details later in the course.)
- Most compilers (including g++) include an implementation of the standard library.

Alexander Stepanov

C++ vs. C

- C++ originated as a C fork, but:
- C is not a subset of C++.

  Some C code does not compile in C++ (or it compiles but then has subtle logical errors).
- Almost no C++ code compiles as C code.

  If it does…then you are not programming with proper C++ !
- More information here: https://en.wikipedia.org/wiki/Compatibility_of_C_and_C%2B%2B

Some useful on-line resources

- **Reference**
  - https://cplusplus.com/reference/
  - https://en.cppreference.com/w/
- **Cheat-sheets**
  - https://github.com/AnthonyCalandra/modern-cpp-features/blob/master/CPP11.md
  - https://github.com/cpp-best-practices/cppbestpractices/blob/master/00-Table_of_Contents.md
- **Advanced**
  - Agner Fog's optimisation guide: https://www.agner.org/optimize/optimizing_cpp.pdf
  - Hardware effects: https://github.com/Kobzol/hardware-effects
- **CppCon** YouTube channel: https://www.youtube.com/@CppCon
- **Performance blogs**
  - https://lemire.me/blog/
  - https://travisdowns.github.io/

Setting-up a development environment

Setting-up a development environment

We suggest you to use the **Linux** operating system.

(Windows treats I/O differently and if you compile your exam project on Windows you may have troubles – it happens to several students every year.)

We will compile and try your project code on Linux anyway, so…

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Some **alternatives**:

- Install Ubuntu on your PC (e.g., dual boot). Different procedures for different PCs.
  Not needed if you have a Mac (unix-based, similar terminal environment).
- Windows Subsystem for Linux (WSL): https://ubuntu.com/tutorials/install-ubuntu-on-wsl2-on-windows-11-with-gui-support#1-overview. **Recommended solution for Windows.**
- Create an Ubuntu virtual machine (https://www.youtube.com/watch?v=x5MhydijWmc).

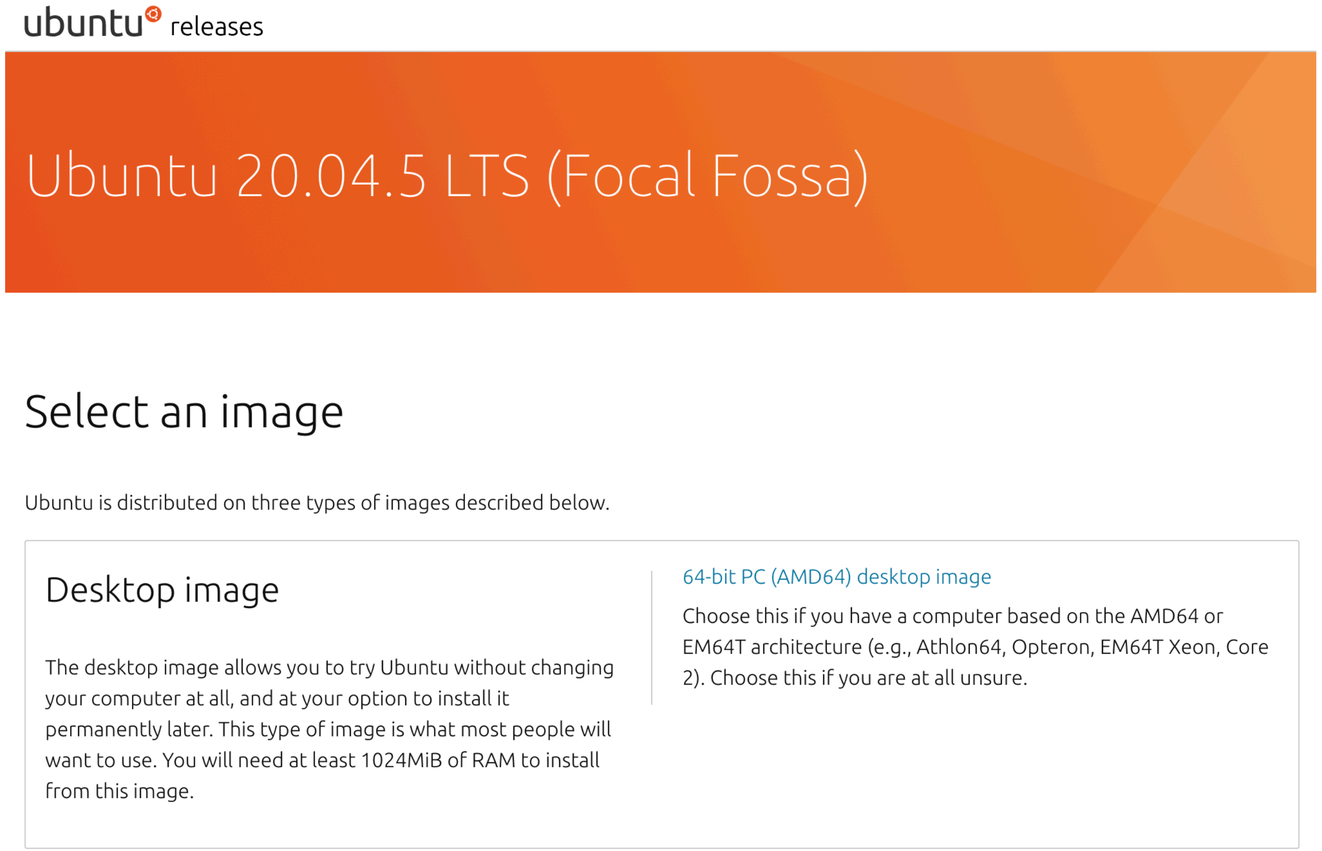
Creating an Ubuntu virtual machine

1. First install Oracle VirtualBox — https://www.virtualbox.org/wiki/Downloads
2. If you use Windows, download “Windows hosts” from the link above (or, if you have a MAC, “OSX hosts”, either Intel or ARM depending on your processor) and execute the file.
3. Proceed with the default installation.
4. Run VirtualBox.
5. Install Ubuntu 20-04 LTS (Focal Fossa) — follow this tutorial: https://linuxhint.com/install_ubuntu_virtualbox_2004/

Install Ubuntu 20-04 LTS (Focal Fossa)

1. Download the iso file (~3.9GB) from http://releases.ubuntu.com/20.04/.

2. When creating a new virtual machine on Virtual Box, specify the downloaded iso file.



Setting-up a development environment

- We will use a minimalist setup. **Abstract away technology and focus on C++ concepts.**
- Two things:

  1. **a text editor** (my preference, Sublime Text, https://www.sublimetext.com/3) to write the code;

  2. **the terminal** to **compile and run** the code.

Setting-up a development environment

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- Two things:

  1. **a text editor** (my preference, Sublime Text, https://www.sublimetext.com/3) to write the code;

  2. **the terminal** to **compile and run** the code.

- A text editor is an application that must be installed first.
- The terminal is another application that is already available in Linux, as well as (usually) the compiler.

Alternatives

- Some alternatives to this setup: configure an **IDE** (Integrated Development Environment).
- Example: Visual Studio Code (VSC, https://code.visualstudio.com/), Eclipse, CLion, etc.
- In class we will **not** discuss how to configure an IDE.

My recommendation

- For this course:

  DO NOT USE ANY IDE.

- I am not against them per se; rather, you have to first learn the hard way.

- They are complex and it's easy to get lost in a configuration madness.

- They do part of the job for you:
  - fixing missing includes,
  - hiding the compilation process, etc.

Terminal: using a Linux shell

Linux shell

- A **shell** is a user interface to **access the operating system**'s services.
- The **Terminal** is an application providing such user interface.
- We will learn how to compile a C++ program from the shell.
- In order to do that, we need some basic notion of the shell itself (mainly filesystem navigation and basic file manipulation).

Graphical vs. textual interface

- All systems have a set of graphical applications (word processor, email reader, internet browser, etc.) that can be controlled using mouse and keyboard.
- All systems can **also** be controlled using a “textual” interface.

Interface	Pros	Cons
Graphical	Easy to learn	Cannot be automatized
Textual	Easy to automatize	Hard to learn

Graphical vs. textual interface

- The **prompt** includes information like the username, the name of the computer, and the current directory.
- Example.

  `user-name@computer-name:~$`

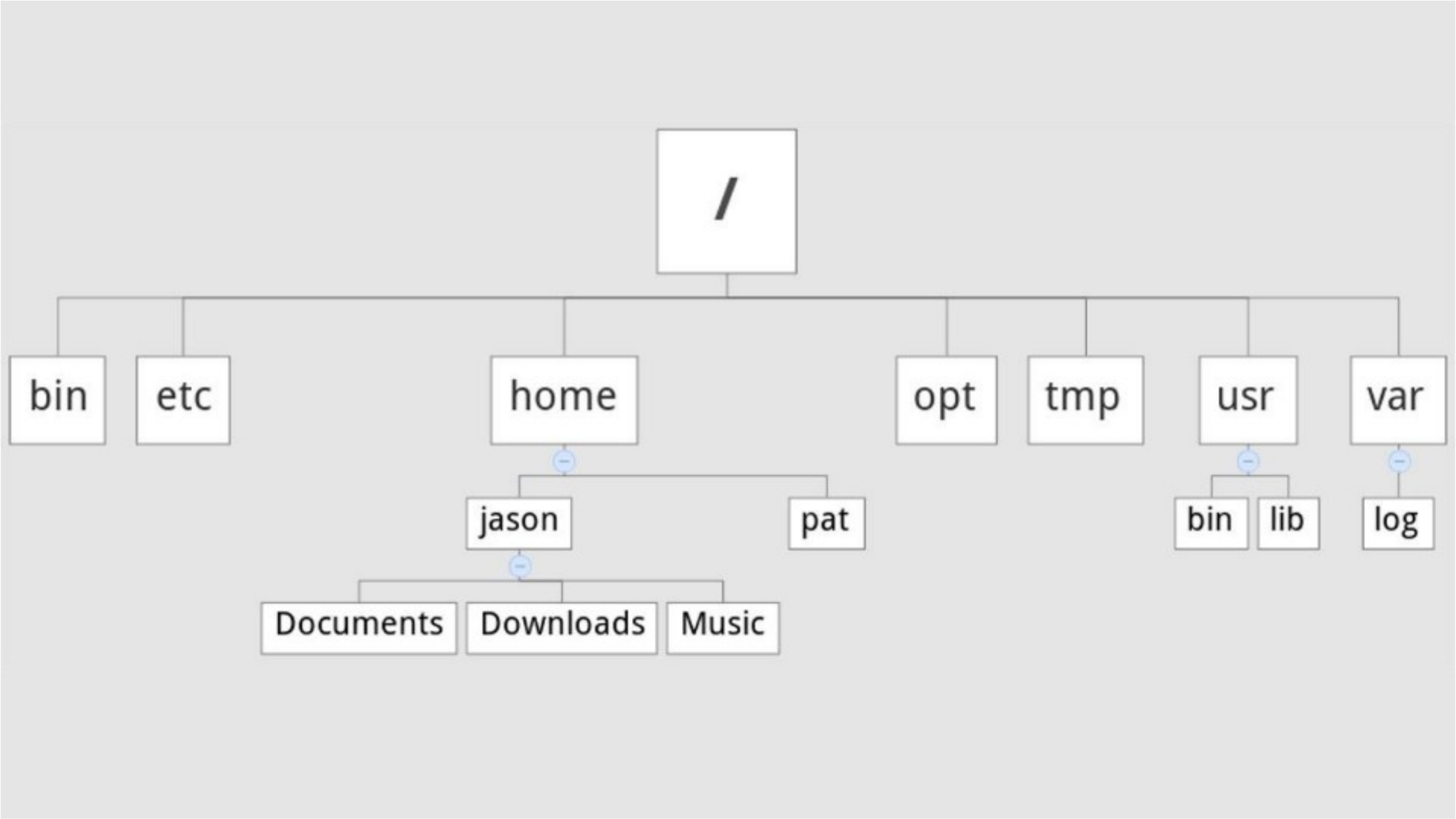
  means:

  - user name → “`user-name`”
  - computer name → “`computer-name`”
  - filesystem position → ~ (“home” directory)
  - Characters @ : $ are separators. To the right of $ we write our commands.

Filesystem

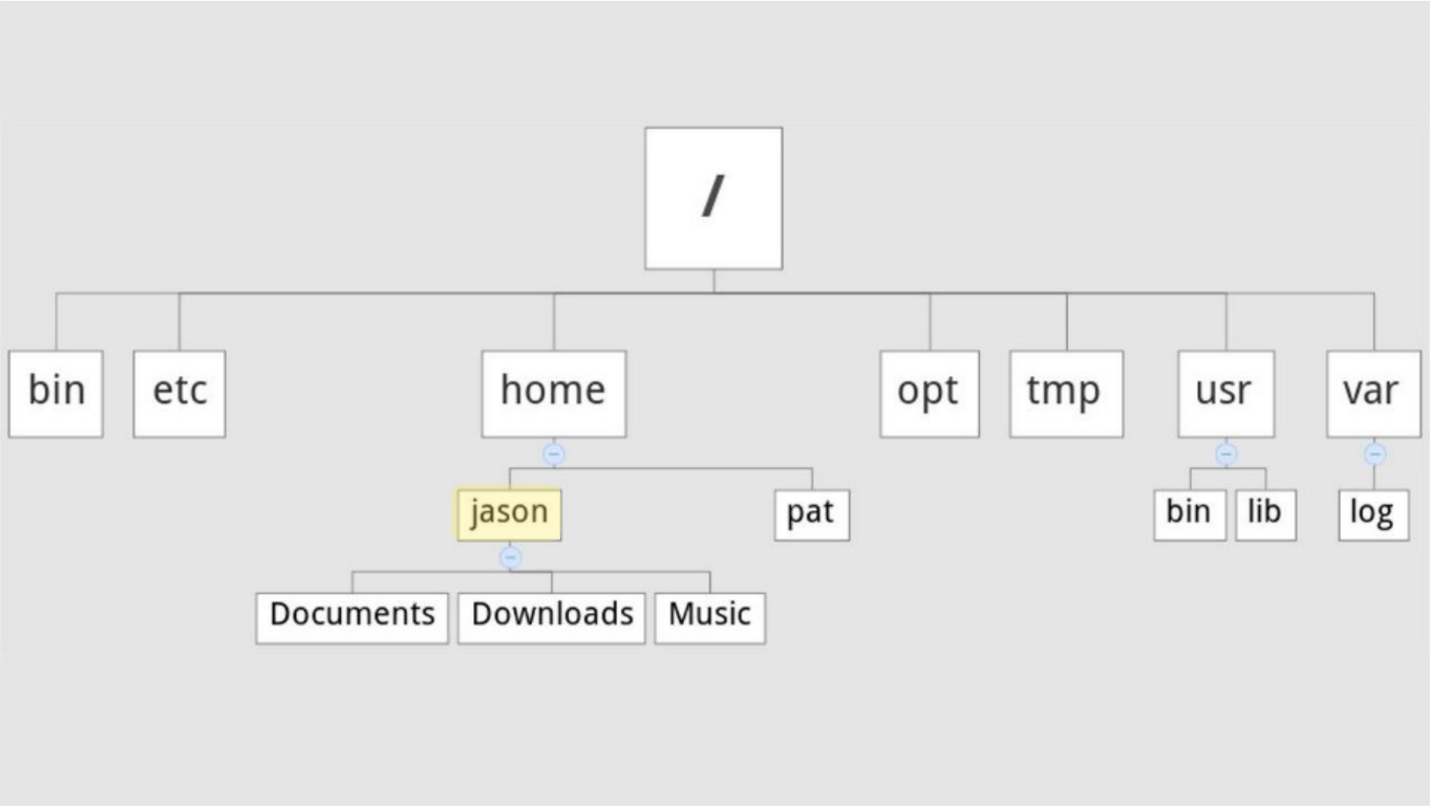
- A “filesystem” is a hierarchic representation (a tree) of a set of files.
- Files are organized into folders (directories).
- Folders can be nested into sub-folders.
- Each file and folder has a name and a path (the path from the root to the object).
- The “root” directory has no name and it is represented as `/` (slash).

Linux filesystem



Linux filesystem

- When opening a new terminal our current working directory is the user's home.
- Command **`pwd`**:
  prints the path of the "working" (current) directory.



Working directory

- The directory where we currently are (shown in the prompt), is called “working directory” or “current directory”.
- By default, the first working directory is the “home” (denoted by the symbol “~”).
- You can see the content of a folder (the list of files and directories) with the command **`ls`** (list).

A simple command: show directory content

- Like all commands in Linux, you can add options to the “`ls`” command to alter its output or influence its behaviour.
- An option is preceded by a dash or a double dash: “`ls -l`” produces a “long format” directory listing; it also shows the permissions, owner, group, size, date and hour of modification.
- “`ls -a`” lists all the files in the directory, including hidden ones (they start with a ‘.’ character).

Man

- **Q.** Doubts on a command or its options?
- Use the manual: **`man`**.
- Example:

```
man ls
```

- Displays the manual entry for “`ls`”.

Moving in the filesystem

You can move the current directory using the “`cd`” command (change directory). Note that prompt changes:

- `cd /home`
- you can move “one directory back” with the command

  `cd ..`

- you always return the home directory with

  `cd`

Run **pwd** to see the current working directory (or look at the prompt).

Absolute vs. relative path

- An **absolute** path starts with a “/” (slash) and specifies the entire sequence of directories from the “root” directory (/) up to the specific file/directory being requested:

  `/home/prog2021/prova`

- A **relative** path does not start with a “/” and it is relative to the current directory:

  `cd folder1`

  works **only if** the folder “`folder1`” exists in the current working directory.

Example

Two ways to go in the directory `/var/local`:

- using the absolute path: `cd /var/local`
- using the relative path:

```
cd /
cd var
cd local
```

Create/delete a directory

- You can create a directory with

```
mkdir dir_name
```

- You can delete an **empty** directory with

```
rmdir dir_name
```

- As a safety measure, the directory must be empty before it can be deleted.
- If the path is not absolute, the new directory is created in the current working directory.

Create/delete files

- You can remove files (but not directories) with

```
rm file1 file2 file3
```

- You can remove files and directory (recursively) with

```
rm -r file1 file2 file3 dir1 dir2
```

- The option **`-f`** forces the operation, skipping warnings.
- Be careful:
  - the files are **DELETED PERMANENTLY** (there is no coming back)
  - with **`-rf`** you can destroy ALL your data

Exercise

- Create a directory “`test`” in your `home` directory.
- Enter in “`test`” directory and create the “`inside`” directory.
- Return in “`test`” directory and remove “`inside`” directory.
- Remove the “`test`” directory.

Command history and tab completion

- It does not take long before the thought of typing the same command over and over becomes unappealing. One solution is to use the **command line history.**

- By scrolling with the [Up] and [Down] arrow keys, you can find your **previously typed commands**.

- Another time-saving tool is known as **command completion**. If you type part of a file or pathname and then press the [Tab] key, the shell presents you with the remaining portion of the available file/path.

  Example: `cd /h`[Tab]

  presents you a list of completions.

Changing name (or moving a file)?

With the command “`mv`” (move) you can:

- rename a file:

```
mv old_filename new_filename
```

- move a file

```
mv /home/prog2021/filename /path/
```

- move AND rename

```
mv /home/prog2021/old_filename /path/new_filename
```

Warning: if the second file exists (`/path/new_filename` in the example), it will be silently overwritten.

Copying files and directories

With the command “`cp`” you can make a copy of a file or a directory:

```
cp old_name new_name

cp file dir_name/

cp old_name dir_name/new_name

cp -r file1 file2 dir1 dir_out/
```

Warning: if the destination file exists, it will be silently overwritten.

Create/display file content

To create a file:

`nano filename` (textual interface)

`gedit filename` (graphical interface on Linux)

To display the contents of the specified file into the screen:

```
less filename
```

```
gedit filename
```

With less, you can use arrows keys and page up/down keys to navigate up and down.
Hit “q” key to quit.

First and last lines

Show the first 10 and last 10 lines:

```
head filename

tail filename
```

Show the first “n” (e.g., 20) and last “n” lines:

```
head -n 20 filename

tail -n 20 filename
```

Concatenate files

Another way to see file contents is to use the `cat` command:

```
cat filename
```

This command displays the entire file, so it is not convenient to use it with big files.

It can also be used to concatenate files:

```
cat file1.txt file2.txt > file3.txt
```

Where “>” redirects the output of `cat` towards the file (instead of standard output), **overwriting its content** (if `file3.txt` already exists, it is overwritten); “>>” appends content instead.

Exercise

- In your home directory, create a new directory called “`exercise`” (`mkdir`)
- Change your directory to the directory `exercise` (`cd`)
- Write your name in the file `name.txt`
- Write your surname in the file `surname.txt`
- Concatenate files `name.txt` and `surname.txt` in the new file `student.txt` (`cat`)
- Visualize the content of the file `student.txt` (`less` or `cat`)

Select lines

To select lines matching a specified “PATTERN” in a file

```
grep PATTERN filename.txt
```

useful to search text in huge files!

Pipelining

The character “|” allows to use the output of a command as input for another program, e.g.:

```
grep pippo filename.txt | head
```

returns the first ten lines of the file `filename.txt` that contain “`pippo`”.

Counting lines

To print the number of lines, words, characters:

```
wc filename
cat filename | wc
```

To print only the lines:

```
wc -l filename
cat filename | wc -l
```

“`wc -c`” counts characters instead; “`wc -w`” counts words.

C++ compiler: g++

g++

- g++ is the compiler that we will use for C++. Usually it is already installed on Linux.
- It is just a call to `gcc` (compiler for C, which you have already used), specialised for C++. Equivalent to:

```
gcc -xc++ -lstdc++ -shared-libgcc
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```

- Basic compilation:

```
g++ myprogram.cpp
```

produces the executable

```
a.out
```

- Option "`–o output`" allows to specify the name of the output:

```
g++ myprogram.cpp –o myprogram
./myprogram
```

Include directives

- In C++ you can **include** `.cpp/.hpp` files with the `include` directive. If you do not specify an **absolute** path, g++ looks for them in the directory containing the `.cpp` file you are compiling.

- **Q.** What if the included files are not in the same directory?

Include directives

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- **Q.** What if the included files are not in the same directory?

- For example you might have a file `foo.hpp` that resides in `/home/prog2021/project/include`.

  To include it in `helloworld.cpp`: write `#include "foo.hpp"` inside `helloworld.cpp` (at the top)

  Compile it with the `-I` option:
  ```
  g++ -I /home/prog2021/project/include/ helloworld.cpp -o helloworld
  ```

- This tells g++ to look for header files specified with `#include` in the directory `/home/prog2021/project/include/`.

Object files

- An object file (usually, with extension `.o`) is the result of the compilation process.
- Object files are source files compiled into binary machine language that may contain (and usually do) references to external functions, defined in other files. Hence, object files have to be “linked” to those references.
- An executable file is the result of linking object files together.

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- Here is the basic idea:

  1. compile each `.cpp` file into a `.o` file using the compiler’s option `-c`;
  2. link the `.o` files (along with libraries, possibly) into an executable.

- Of course, one of these `.cpp` files must define `main()`.

Object files

- Suppose we have `main.cpp`, `foo.cpp` and `bar.cpp` and want to create an executable `fubar`:

```
g++ -c foo.cpp -o foo.o

g++ -c main.cpp -o main.o

g++ -c bar.cpp -o bar.o

g++ foo.o main.o bar.o -o fubar
```

- The first three commands generate `foo.o`, `main.o` and `bar.o` respectively (option `–c`: does not generate executables, only object files).

- The last command links them together (no option `–c`) to generate the executable file.

Common compiler flags

`-g` turn on debugging. Needed for the debugger/profiler (e.g., gdb and Valgrind — next lectures)

`-Wall` and **`-Wextra`** turn on most warnings

`-O` turn on optimizations (capital “O”, next slide)

`-o` `<name>` name of the output file

`-c` output an object file (.o) rather than an executable file

`-I <include path>` specify an include directory

`-L <library path>` specify where libraries are located

`-l <library>` link with library `lib<library>.a` (esempio: `-lm`)

`-std=c++17` specifies the version of the C++ standard to use.

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We will **not** use them in this course.

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Compiler optimization flags

option	optimization level	execution	code	memory	compile time
`-O0`	optimization for compilation time (default)	+	+	-	-
`-O1`	optimization level 1 for execution time	-	-	+	+
`-O2`	optimization level 2 for execution time	--		+	++
`-O3`	optimization level 3 for execution time	---		+	+++
`-Os`	optimization for code size		--		++
`-Ofast`	03 with fast accurate math calculations	---		+	+++

Need help?

Don't tell me you climbed **all the way up here** before trying **Google??**
Pirillo & Fitz
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